



**Verified Carbon
Standard**

TABREED DISTRICT COOLING GROUP PROJECT



Project title	TABREED DISTRICT COOLING GROUP PROJECT
Project ID	4519
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1 PROJECT DETAILS

1.1 Summary Description of the Project

National Central Cooling Company PJSC (TABREED) is implementing highly efficient district cooling project titled 'TABREED DISTRICT COOLING GROUP PROJECT (hereafter called “Project” or “Grouped Project”)’. The project will reduce electricity consumption by providing highly efficient district cooling solution (air conditioning) to meet the cooling demand for residential and commercial consumers in the Abu Dhabi Emirates, United Arab Emirates (UAE). The highly efficient district cooling will displace the cooling from conventional/less efficient cooling system and reduce the electricity consumption which would result in reduction in greenhouse gas (GHG) emissions.

In the UAE air conditioning is necessity to survive. The penetration of cooling technology is almost 100%. Most of the residential and commercial consumers use window air conditioner or split air conditioner to meet the cooling demand. The commercial consumers have started shifting towards air cooled centralized cooling. These technologies are less efficient with respect to district cooling.

The purpose of the project instance 1 is to introduce new district cooling plant to supply cooling to residential and commercial consumers in the Sea World, Yas Island Abu Dhabi, United Arab Emirates. The Project will replace isolated and less efficient cooling technologies (air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO₂) emissions. This is estimated that the district cooling will save more than 40-50% of electricity with respect to the baseline technology.

The grouped project will install district cooling plants of various capacities to residential and commercial consumers who would have used the less efficient air-cooled reciprocating chiller system. All the project instances to be included in this grouped project will be from within Abu Dhabi Emirates, United Arab Emirates.

The first project activity instance is installed in the sea world Abu Dhabi with a capacity of 9000 TR. The estimates of annual average GHG emission reductions of project activity instance 1 is 19,320 tCO₂e/annum

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	05-Jan-2024	VCS	KBS Certification Services Limited	One year

1.3 Sectoral Scope and Project Type

Complete the table below with information relevant for non-AFOLU projects:

Sectoral scope ¹	01 Energy industries (renewable - / non-renewable sources)
Project activity type	Grouped project activity

Complete the table below with information relevant for AFOLU projects:

Sectoral scope	Not applicable.
AFOLU project category ²	
Project activity type	

1.4 Project Eligibility

1.4.1 General eligibility

The scope of the VCS Program includes:

- The Grouped Project activity involves replacement of less efficient cooling with more energy efficient district cooling. The baseline & monitoring methodology applied for this project is “UNFCCC CDM Methodology AM0117: Introduction of a new district cooling system; Version 02”, which is large scale methodology.
- The six Kyoto Protocol greenhouse gases are considered under the VCS program and project is reducing the emissions of CO₂ mainly due to reduction of electricity consumption for cooling.
- Ozone depleting substances: NA
- Project activities supported by a methodology approved under the VCS Program through the methodology approval process: NA
- Project activities supported by a methodology approved under a VCS approved GHG program unless explicitly excluded under the terms of Verra approval: The methodology AM0117 (version 02.0) is approved under CDM Program, which is a VCS approved GHG program.

The project does not belong to the projects excluded in Table 1 of VCS Standard 4.5. Thus, the project is eligible under the scope of the VCS program.

1.4.2 AFOLU project eligibility

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

Not applicable as the project is Non-AFOLU Project.

1.4.3 Transfer project eligibility

Not applicable as the project is not a transfer project.

1.5 Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

The project has been designed as a grouped project having multiple project activity instances.

Eligibility Criteria

The project is a grouped project and following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity:

As per VCS Standard v 4.5 section 3.6.1, methodology AM0117 is applicable for program of activities (section 5.8 of AM0117) and therefore multiple project activity requirements of the VCS are followed.

Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AM 0117 version 02) as defined in section 3.2.

Geographical Area: The project activity instance locates in the geographic boundary for the grouped project, i.e., Abu Dhabi Emirates, and owned by the project owner or the subsidiary of the project owner. As per section 3.11 of the VCS standard a geodetic coordinate for each instance, will be provided in a KML file.

Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in the respective methodology. The baseline scenario is based on technology penetration and therefore latest/updated third party report shall be provided for every new project instance inclusion.

Technology type: The project activity instances to be included in the grouped project activity is the implementation of new district cooling plant with or without thermal energy storage.

Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology.

Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.

Capacity limits: No capacity limits applied in the methodology and therefore no capacity limit will be applicability for inclusion of project activity instances.

Stakeholder consultation: Before inclusion of any new project instance stakeholder consultation will be conducted.

Environmental Impact: Every new project instance will follow the host country's environmental regulations related with environmental approvals.

Sampling: No sampling will be used for the emission reduction calculation. All the project instances shall have the monitoring parameters as explained in monitoring section of the methodology AM0117 version 02.

Unique Identification: All the new project instance will be named uniquely based on location as the PI 1 is called 'Sea World District Cooling Plant'.

Programme management, collection of data and record keeping: Project proponent Tabreed will be responsible for keeping all the emission reduction data, including the details of project instances start date, commissioning document and other required information.

The eligibility criteria are presented in the table below:

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion
1	Applicability Conditions of the applied methodology.	The project activity instances shall meet applicability conditions for applicable methodology (AM 0117 version 02) as defined in section 2.2.	The justification shall be part of 'Monitoring Report' during the first-time inclusion of any new instance.
2	Geographical Area	The project activity instances for the grouped project will be located in the geographic boundary of Abu Dhabi Emirates and owned by the project owner or the subsidiary of the project owner.	KML file, GPS Coordinates, maps as per technical reports, proposals and feasibility studies, Sales documents, commissioning and approval documents.
3	Baseline scenario	All Project Activity Instances shall meet the baseline definition as defined in the respective methodology. The baseline	Latest/updated third party report shall be provided for every new project instance inclusion.

		scenario is based on technology penetration.	
4	Technology type:	The project activity instances to be included in the grouped project activity is the implementation of new district cooling plant with or without thermal energy storage. Each instance will exclusively serve new buildings similar to instance 1. Any of the project instances will not supply to any existing building (buildings within the project boundary that were connected to a baseline cooling technology before the start of the project activity and are supplied district cooling by the project activity).	Technical reports, proposals and feasibility studies Cooling demand Lay-out diagram for the DCP. Confirmation by PP regarding applicable Project Scenario for the Project activity. Purchase order of equipment.
5	Additionality	The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology. Additionality for this instance has been demonstrated by using requirements provided in para 5.2.1; Step 1: Penetration and Performance assessment, in the applied methodology. a. The seasonal energy efficiency ratio (SEER) of the DCP involved in the project activity is higher than a benchmark SEER and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark boundary in terms of cooling output.	The seasonal energy efficiency ratio (SEER) calculations. The share of technologies can be derived from official country reports, third party surveys and/or credible international sources (e.g. International Energy Agency).

		The future instances shall comply above requirement of methodology for additionality demonstration.	
6	Start date	The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The evidence to support the start date of future instances shall be furnished to VVB.
7	Capacity Limits	No capacity limits are mentioned in the applied methodology and therefore no capacity limit will be applied for inclusion of new project activity instances.	Not applicable
8	Stakeholder consultation:	The local stakeholder consultation will be conducted before inclusion of any new project instance.	Meeting notices; Minutes of meeting; Documents relating to Feedback / comments redressal.
9	Environmental Impact:	Every new project instance will follow the host country's environmental regulations related with environmental approvals	Environmental Impact study, if applicable.
10	Sampling:	No sampling is used for the emission reduction calculation. All the project instances shall have the monitoring parameters as explained in monitoring section of the methodology AM0117 version 02	Not applicable
11	Conditions to avoid double counting - Unique Identification:	All the new project instance will be named uniquely based on location as the PI 1 is called 'Sea World District Cooling Plant'. To prevent double counting of District Cooling Plants under the project, District cooling plant shall be uniquely marked, and new project instance implementers shall provide assurances/declarations	PI title, GPS coordinates, technical reports, proposals and feasibility studies. Analysis of projects in VCS/CDM pipeline.

		that they are not included in any other CDM or Voluntary GHG offset schemes outside of this project.	
12	<u>Programme management-</u> Approval of new project instance by project proponent; <u>collection of data and record keeping:</u>	The project proponent approves each project instance to be included into its registered group project activity. Project proponent Tabreed will be responsible for keeping all the emission reduction data, including the details of project instances start date, commissioning document and other required information.	Statement of project proponent giving approval for the new project instance to be included into its registered group project activity.
13	Conditions to provide an affirmation that funding from Annex I parties, if any, does not result in a diversion of official development assistance.	under this project, new project instance operators shall be required to provide an affirmation that funding from annex I parties, if any, does not result in a diversion of official development assistance	Individual new project instance operators shall sign ODA declarations for new project instances implemented by them to attest that no funding from annex 1 parties that leads to diversion of ODA has been used in the project. the statement shall specify whether or not annex 1 funding is used within the project where funding is sought from annex 1 parties the ODA declaration shall be supported by details regarding annex 1 funding that affirms that the funding doesn't divert ODA For this instance, no funding from Annex-1 parties is received declaration to this effect has been furnished to VVB.
14	VCU ownership	End users receiving Cooling demand under the specific new project instance contractually cede their rights to claim and own emission reductions under the VCS to the project proponent of the group project.	A default agreement for end users including the provision that emission reductions generated by the DCP are owned by the project proponent, will be

			provided for each new project instance. Sample agreement with end user for Waiver of carbon credits are submitted to the VVB for the first project instance.
15	Awareness and agreement of those operating a new project instance on group project subscription	Contractual provisions to ensure that those operating the new project instance are aware and have agreed that their activity is being subscribed to the new project instance.	Evidence for inclusion, in case new project instance operators are different from the project proponent: a declaration from project proponent/implementer, stating that they are aware and have agreed that their activity is being subscribed to the group project will be provided for each new project instance.

1.6 Project Proponent

Organization name	NATIONAL CENTRAL COOLING COMPANY PJSC
Contact person	Antonio Di Cecca
Title	Chief Operating Officer
Address	PO Box 29478, Abu Dhabi, United Arab Emirates
Telephone	T + 971 2 202 0400 Ext 434
Email	adicecca@tabreed.ae

1.7 Other Entities Involved in the Project

Not applicable

Organization name	
Role in the project	
Contact person	
Title	

Address	
Telephone	
Email	<i>The email address domain must match that of the organization.</i>

1.8 Ownership

The respective project commissioning certificate for project activity are the supporting documents to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.5) “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. Also, other legal compliances may be considered;

- Consent to Operate
- Commissioning documents

1.9 Project Start Date

Project start date	25-May-2023
Justification	As per section 3.8 of the VCS Standard (v4.5), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals. The project activity is a non-AFOLU project and the date of commissioning of first Project is 25-May-2023. Thus, the project start date is as 25-May-2023.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	25-May-2023 to 24-May-2030

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
25-May-2023 to 31-December-2023	11,681
01-January-2024 to 31-December-2024	19,323
01-January-2025 to 31-December-2025	19,323
01-January-2026 to 31-December-2026	19,323
01-January-2027 to 31-December-2027	19,323
01-January-2028 to 31-December-2028	19,323
01-January-2029 to 31-December-2029	19,323
01-January-2030 to 24-May-2030	7,623
Total estimated ERRs during the first or fixed crediting period	135,242
Total number of years	7
Average annual ERRs	19,320

1.12 Description of the Project Activity

The Project Instance 1 is the introduction of a new district cooling plant with a state-of-the-art thermal energy storage system to supply cooling to residential and commercial consumers in sea world, Yas island, Abu Dhabi.

The Project Activity includes the erection of a new district cooling plant of 9000 RT. The other project instances will be included in future, which may include District cooling plants with or without Thermal Energy Storage.

Plant Capacity Configurations			
Project	Number of chillers	Unit capacity (RT)	Total capacity (RT)
Instance 1	3	3000	9000

Technical information on key equipment employed by the proposed Project Activity is described in the following.

District Cooling Plant

The new district cooling plant will employ water cooled 3.3 kV electric driven centrifugal chillers with thermal energy storage and associated auxiliary equipment like cooling towers and pumps for the production and distribution of chilled water to various building via a network of underground insulated primary piping system.

The project district cooling plant capacity is 9,000 Refrigeration Tones (RT). The district cooling plant capacity is achieved through the use of centrifugal chillers. The district cooling plant will serve new buildings.

Thermal Energy Storage (TES) systems provide the ability to store cooling energy during economically more viable periods for re-use during the period of higher costs. Other advantages to use Thermal Energy Storage systems are that lower nighttime temperatures allow refrigeration equipment to operate more efficiently than during the day, reducing energy consumption. With TES less chiller capacity is required, which means lower capital equipment costs, use of off-peak electricity to store energy for use during peak hours and daytime peaks of power consumption is reduced, avoiding the need for expensive power capacity additions and associated emissions. The TES will be included in the future project instances as and when applicable.

The district cooling plant will use 3.3 kV electrical-driven centrifugal chillers with R-134a or equivalent environment-friendly refrigerant with zero ozone depletion potential. The proposed chillers can deliver a cooling capacity of 3,000 TR at zero tolerance. A total of three chillers have been selected for the current phase. The cooling capacity and operating conditions of the chiller evaporator / condenser side are as indicated in the following schedule.

Chiller Cooling Capacity	Chilled Water Temperature		Condenser Water Temperature	
TR	Supply (°C)/(°F)	Return (°C)/(°F)	Supply (°C)/(°F)	Return (°C)/(°F)
3,000	4.44/40	13.33/56	35/95	40.56/105

The district cooling plant will supply chilled water at 4.44 C. The return temperature is 13.33 C.

The table below presents the technical specification of the district cooling plant for phase 1:

Unit Description	Unit capacity		Duty	Total capacity	
Chiller	3,000 (10,563)	TR (KW)	3	9,000 (31,690)	TR (KW)

Cooling Tower	9,150 (577.3)	GPM (L/s)	3	27,450 (1731.9)	GPM (L/s)
Primary Chilled Water Pumps	6,000 (378.5)	GPM (L/s)	3	18,000 (1135.5)	GPM (L/s)
Secondary Chilled Water Pumps	4,875 (307.6)	GPM (L/s)	4	19,500 (1230.4)	GPM (L/s)
Condenser Pump	9,150 (577.3)	GPM (L/s)	3	27,450 (1731.9)	GPM (L/s)

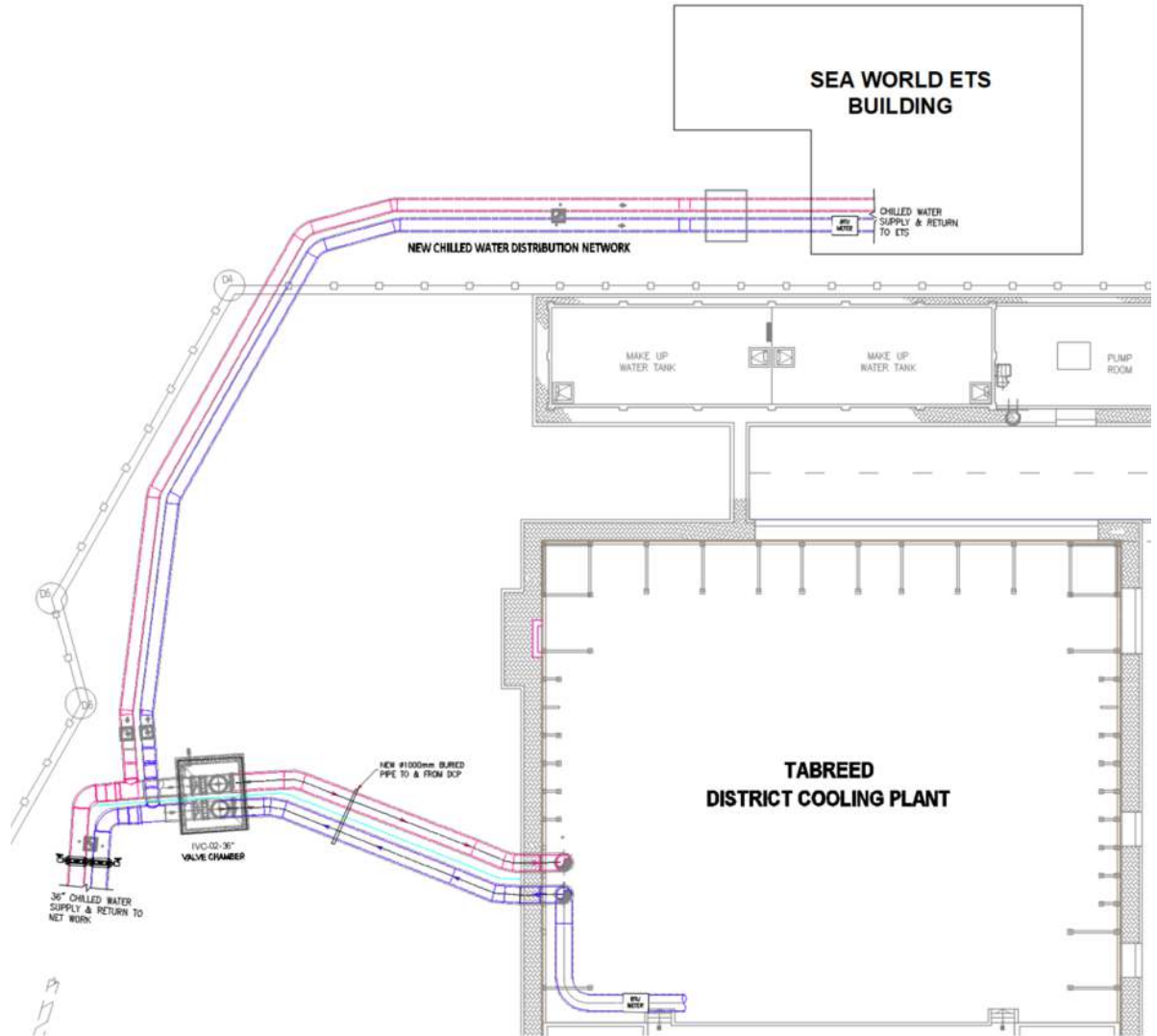
Expected seasonal energy efficiency ratio: 0.84 kW/TR

The new primary district cooling network will be developed for new buildings. There is an already existing primary district cooling network in Yas Island, project will be connecting to that network as well as standby.

For project instance 1; the dedicated distribution network is developed from district cooling plant to the sea world building as presented in the diagram below:

The district cooling plant started operation during May 2023 at the similar time when the sea world buildings³ are opened.

³ <https://www.seaworldabudhabi.com/en/news/grand-opening-22-may-2023>



The similar type of arrangement will be followed for all the future instances.

This connection to existing network is for managing the redundancy, in case of any operational challenges with the district cooling plant.

The district cooling project is developed for the new development/new buildings and the distribution for the district cooling chilled water will also be developed.

1.13 Project Location

The grouped project will be located in Abu Dhabi Emirates in United Arab Emirates. The grouped project covers Abu Dhabi Emirates boundary as presented below⁴:

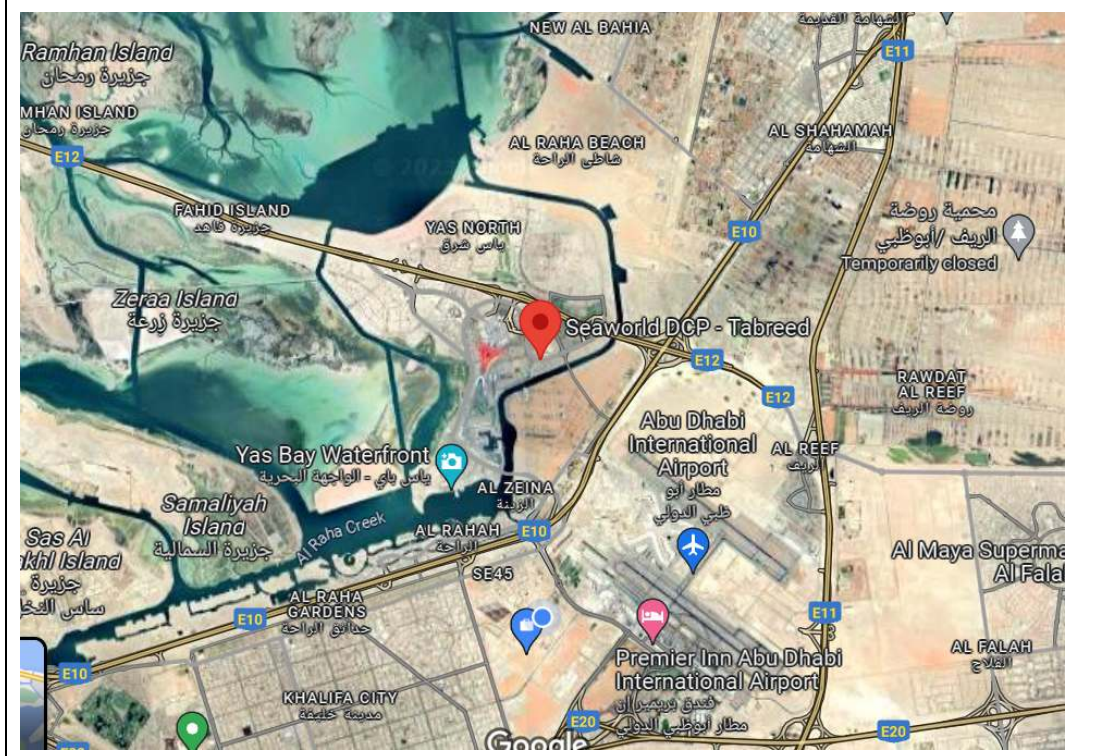
Latitude: 22°40' N to 25° North

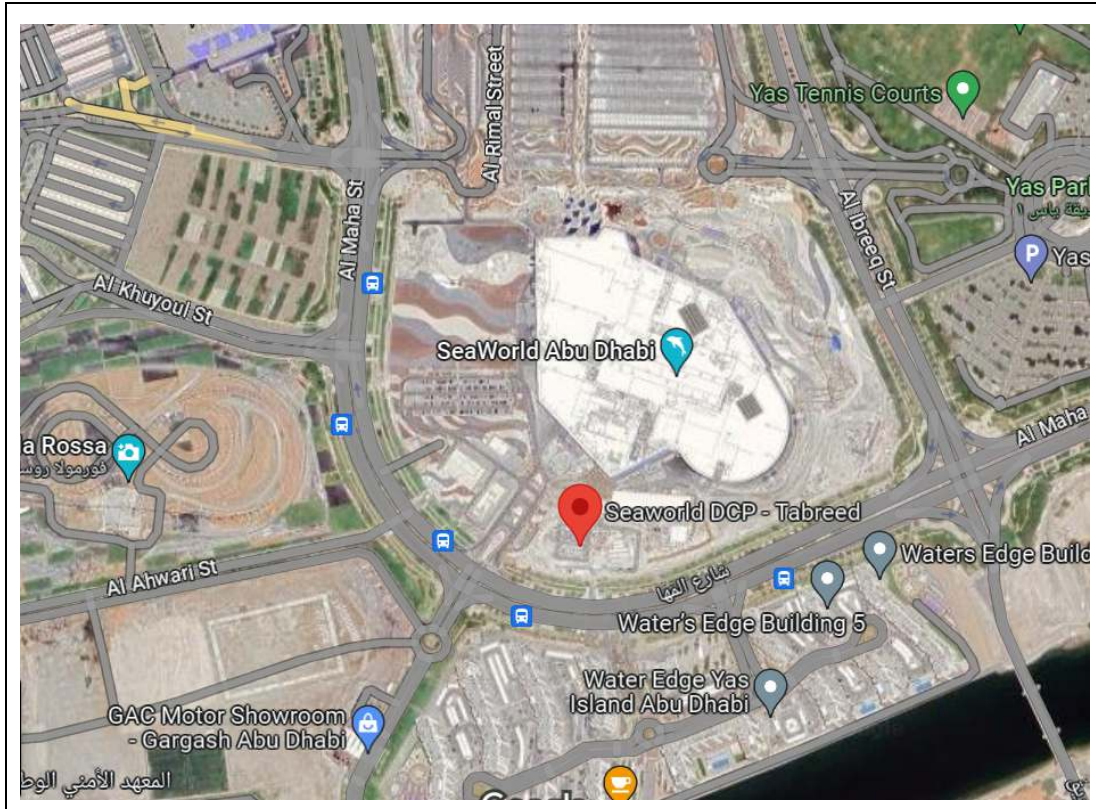
Longitude: 51° to 56° East

The project activity instance 1 is installed in Sea World Abu Dhabi with following coordinates:

Facility Name	Sea World District Cooling Plant
Facility Activity	District Cooling Plant
Plot Area	10099.95-meter square
Location	Yas Island Sector YS6
Plot number	P10
Latitude	24.483103977 N
Longitude	54.618530045 E

⁴ https://en.wikipedia.org/wiki/Emirate_of_Abu_Dhabi





1.14 Conditions Prior to Project Initiation

The baseline scenarios are the same as the conditions existing prior to the project initiation. The details are showed in the Section 3.4 Baseline Scenario.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project complies with all applicable laws. The main laws are:

1. Environmental laws of Abu Dhabi: The project has already received the environmental permit for construction and operation.
2. District cooling regulatory framework under Department of Energy (DoE)⁵: Under Law No 11 of 2018, the Department of Energy (DoE) has a duty to regulate the central / district cooling sector in the Emirate of Abu Dhabi. Following extensive consultation with the public and private stakeholders, the DOE has launched the District Cooling regulations upon the government approval, comprising technical, legal and economic dimensions.

⁵ <https://www.doe.gov.ae/en/Legislation-and-Compliance/Economic-Regulations/District-Cooling>

The economic dimension of this framework will look into aspects of market competition and price regulation, to ensure that:

- There is increased fairness and transparency in the market as well as enhance service levels for end-users.
 - The price for new district cooling schemes is fair and lower than the cost of the best alternative available for cooling.
 - The benefits of district cooling are properly reflected in end-users tariffs.
 - District cooling providers can recover their efficient costs and sector profitability is fostered.
 - The price control process is easy to implement and sustainable in the long term.
3. The two technical regulations have been published and project is in compliance with these codes:
- a. District Cooling Technical Code dated 15/03/2020
 - b. District Cooling Metering Code dated 15/03/2020

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The proposed Project Activity contributes to sustainable development in the United Arab Emirates by the following means:

Economic indicators

The Project Instance contributes to the

1. Conservation of energy sources through a more efficient way of generation and distribution of cooling energy, thus allowing for decreased electricity consumption per ton of cooling supplied. This contributes to UN SDG indicator 11 'sustainable cities' Target 11.6: "Reduce the environmental impacts of cities".
2. Education and training of staff in the operation of the district cooling system. In addition, staff will be trained in the application of the international VCS standards of the Project Instance Activity monitoring and quality assurance / quality control.

Social indicators

The Project Instance contributes to the positive quality of life and benefits the local community by helping ensure a less energy intensive generation and supply of cooling energy.

Environmental indicators

The Project Instance

1. Will be reducing an estimated total of 19,320 tCO₂ of emission reductions per year and therefore contributing to UN SDG 13 'Climate Action'.
2. Complies with all applicable national environmental regulations and standards and has no impact on air, water, soil or groundwater pollution.
3. will be reducing the usage of desalinated water as it allows for the future introduction of grey water (treated sewage water) to be used as cooling medium.

Technical indicators

The Project Instance introduces a new and energy efficient technology and know-how into the United Arab Emirates, which can successfully be replicated to other cities within the region. The Project Instance will contribute to the increase in utilization and application know-how of this type of technology (district cooling).

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Leakage emissions are considered as per the applicable methodology.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

There are no information or incidents that will have bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The project is an energy efficiency project. The stakeholders are: 1. Employees 2. Government/Competent Authority a. Department of Energy b. Abu Dhabi Distribution Company (ADDC) c. Environmental Agency Abu Dhabi (EAD) d. Abu Dhabi Department of Municipalities and Transport (DMT) 3. Local Communities 4. Non-Governmental Organisation (NGO)/Community Based Organisation (CBO)
Legal or customary tenure/access rights	Not applicable
Stakeholder diversity and changes over time	The stakeholders are diverse and no changes are expected over time.
Expected changes in well-being	Other than emission reductions no other changes in the well-being of stakeholders.

Location of stakeholders	Stakeholders are diverse and located in the emirates of Abu Dhabi.
Location of resources	Not applicable.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	19-September-2023
Stakeholder engagement process	The stakeholder consultation had been conducted on 19th September 2023, where various stakeholder attended the consultation meeting. The stakeholder meeting was recorded.
Consultation outcome	The stakeholders enquired about the benchmark, benefits of district cooling systems, refrigerant usage and source of water. The stakeholders were very supporting of the project.
Ongoing communication	Tabreed has a customer service charter (https://www.tabreed.ae/customer/) which provides the detailed process on how the complaints are recorded and addressed. The main extract from the charter: 5- Complaints Handling Process 5.1- Lodging a Complaint End-user Customers can lodge complaints by calling 800 TASLEEM (8275336), emailing customerservice@tasleem.ae, or visiting the Tasleem website: www.tasleem.ae. Bulk-offtake customers will be able to log complaints by calling 800 Tabreed (8227333), sending an email to customerservice@tabreed.ae, or contacting their designated Account Manager. The complaint can be assigned, tracked, and monitored by the Customer Service team leads. When a call type is selected in the billing system, the operator has the option to list the complaint. All complaint types are included in the complaint reports.

	5.2- Receipt and Acknowledgment of Complaint Customers will receive an acknowledgment receipt of their complaint via either a phone call or an e-mail. This acknowledgement is made on the day the complaint is lodged, provided it is a business day, or the following business day if the complaint was made during the weekend or on a public holiday. 5.3- Accessibility to the Complaint End-user Customers can request copies of their complaint log by emailing customerservice@tasleem.ae or calling 800 TASLEEM (8275336) during normal business hours. Bulk-offtake Customers can request copies of their complaint log by emailing customerservice@tabreed.ae or calling 800 TABREED (8227333) during normal business hours
Stakeholder input	All the inputs were considered and responded accordingly. The stakeholder inputs process is continuous process and the comments will be recorded and appropriately responded.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Project has received all the required consents from government authorities.
Outcome of FPIC	Project has received all the required consents from government authorities. Project has not encroached on land and relocated people.

2.1.4 Grievance Redress Procedure

Development process	The project owner has operating complaint procedure. The same is used for the project.
Grievance redress procedure	Tabreed has a customer service charter (https://www.tabreed.ae/customer/) which provides the detailed

process on how the complaints are recorded and addressed. The main extract from the charter:

5- Complaints Handling Process

5.1- Lodging a Complaint

End-user Customers can lodge complaints by calling 800 TASLEEM (8275336), emailing customerservice@tasleem.ae, or visiting the Tasleem website: www.tasleem.ae.

Bulk-offtake customers will be able to log complaints by calling 800 Tabreed (8227333), sending an email to customerservice@tabreed.ae, or contacting their designated Account Manager.

The complaint can be assigned, tracked, and monitored by the Customer Service team leads. When a call type is selected in the billing system, the operator has the option to list the complaint. All complaint types are included in the complaint reports.

5.2- Receipt and Acknowledgment of Complaint

Customers will receive an acknowledgment receipt of their complaint via either a phone call or an e-mail. This acknowledgement is made on the day the complaint is lodged, provided it is a business day, or the following business day if the complaint was made during the weekend or on a public holiday.

5.3- Accessibility to the Complaint

End-user Customers can request copies of their complaint log by emailing customerservice@tasleem.ae or calling 800 TASLEEM (8275336) during normal business hours.

Bulk-offtake Customers can request copies of their complaint log by emailing customerservice@tabreed.ae or calling 800 TABREED (8227333) during normal business hours

2.1.5 Public Comments

Comments received	Actions taken
No public comments have been received for the project	

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risk identified	Project is energy efficiency and operation is limited to expert operators and therefore no risk to stakeholder participation.
Working conditions	No risk identified	The project owner is listed public company and abide by all the legal requirements.
Safety of women and girls	No risk identified	The service provided by project is common to all irrespective of gender and therefore no risk identified.
Safety of minority and marginalized groups, including children	No risk identified	The service provided by project is common to all irrespective of gender and therefore no risk identified.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risk identified	The project is housed in dedicated building and following all the environmental approvals and therefore no risk identified.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	In UAE labour laws prohibit any type of discrimination and harassment. Federal Decree-Law No. 33/2021, known as the Labour law, expressly addresses discrimination in various aspects, emphasizing the importance of treating employees fairly and equally. Article 4 of the Labour Code prohibits discrimination based on race, colour, sex, religion, national origin, social origin, or disability. Project owner is listed company and abiding by all the applicable law and therefore no discrimination and sexual harassment occurred and will occur in the future.
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Management experience	The project owner has been implementing similar projects in last 25 years and have sufficient experience on managing all the requirements.
Gender equity in labor and work	Tabreed believes in gender equality and publicly publishing the ESG related information. https://www.tabreed.ae/wp-content/uploads/2023/07/Tabreed_ESG_Report_2022.pdf
Human trafficking, forced labor, and child labor	In UAE laws prohibit any type of human trafficking, forced labor and child labor. The project owner is abiding by all the applicable laws.

2.3.2 Human Rights

Tabreed respects internationally recognised human rights in its relations with public authorities. Tabreed is committed to adhering to international laws and conventions, including, but not limited to, the International Labour Organization (ILO) core conventions: Conventions No. 29 and No. 105 on the Abolition of Forced Labour; Conventions No. 138 and No. 182 on the Abolition of Child Labour; and Conventions No. 100 and No. 111 on Non-discrimination.

More information available on https://www.tabreed.ae/wp-content/uploads/2023/07/Tabreed_ESG_Report_2022.pdf

2.3.3 Indigenous Peoples and Cultural Heritage

The project has received all the approvals for the structures of the building from the respective authorities. This approval confirms that all the required regulations have been followed.

2.3.4 Property Rights

Rights to territories and resources	Project owner has all the rights on the project. The land has been leased and the construction has been carried out by project owner.
Respect for property rights	Not applicable.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable
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Summary of the benefit sharing plan	Not applicable
Approval and dissemination of benefit sharing plan	Not applicable

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	The land used was already developed and no impact on biodiversity due to project.
Soil degradation and soil erosion	No risk identified	The land used was already developed and no impact on biodiversity due to project.
Water consumption and stress	No risk identified	The project will be consuming water. The water will be coming from sea water desalination plant. No impact on potable water.
Usage of fertilizers	No risk identified	No use of fertilizer.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species and habitat	Not applicable
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2.4.2 Introduction of species

N/A

2.4.3 Ecosystem conversion

N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AM0117	AM0117: Introduction of a new district cooling system	2.0
Tool	Tool01	Tool for the demonstration and assessment of additionality	7.0
Tool	Tool03	Tool to calculate project or leakage CO2 emissions from fossil fuel combustion	3.0
Tool	Tool05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	3.0
Tool	Tool07	Tool to calculate the emission factor for an electricity system	7.0
Tool	Tool11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1
Tool	Tool23	Additionality of first-of-its-kind project activities	3.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AM0117	The methodology is applicable to project activities that reduce CO ₂ emissions by means of one, or a combination, of the following measures:	Applicable The Project is new district cooling system that supply cooling to residential and

	(a) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through a new dedicated distribution network; (b) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through an existing dedicated distribution network; (c) Expansion of the existing district cooling system(s) by adding a new district cooling plant(s) with or without expanding a dedicated distribution network.	commercial consumers through dedicated distribution network. In the second phase of the Project, it will be connected with the existing dedicated distribution network.
AM0117	Emission reductions that are gained due to the switch of the energy sources shall not be claimed by applying this methodology alone. Therefore, emission reductions due to displacement of the baseline power source can be claimed by the means of the application of this methodology in combination with another relevant approved methodology. In doing so, interactive effects shall be considered as per the “Guidelines for the consideration of interactive effects for the application of multiple CDM methodologies for a programme of activities”. For example, if the project district cooling plant is powered partially or completely by a dedicated renewable energy power plant, a project proponent may wish to consider small scale methodology ‘AMS-I.F: Renewable electricity generation for captive use and mini-grid’ to account for emission	Not applicable The electricity from the grid is the power source in the baseline scenario as well as in the project instance and therefore project will not replace the energy source and therefore this condition is not applicable.

	reductions due to shift of the power source. Otherwise, the most conservative electricity emission factor should be applied to determine emission reductions.	
Tool01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Not applicable The Project is using the approved methodology and using the additionality as per the approved methodology.
Tool01	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Not applicable Methodology provided the specific additionality procure and additionality tool is provided as an option.
Tool03	This tool provides procedures to calculate project and/or leakage CO ₂ emissions from the combustion of fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	Applicable The project involves calculation of CO ₂ emissions from fossil fuel combustion based on the quantity of fuel combusted and its properties.
Tool05	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:	Applicable The project consumes electricity from grid, which belongs to Scenario A.

	(a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
Tool07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in	Not Applicable Due to unavailability of the data for calculation of OM, BM and CM, project owner is requesting a deviation to use publicly available grid emission factor for

	savings of electricity that would have been provided by the grid (e.g. demand side energy efficiency projects).	the project. Source of the data is described in detail in the deviation section.
Tool07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off grid power plants. In the latter case, two sub options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off grid power generation” should be met. Namely, the total capacity of off grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Not Applicable
Tool07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	This condition is not relevant, there is no part of the power system of this project located in Annex I countries.
Tool07	Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not Applicable
Tool11	This tool provides a stepwise procedure to assess the continued	Applicable

	validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate whether the current baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period.	The Project will use this tool during renewal of crediting period.
Tool23	This methodological tool is applicable to project activities that wish to use the "first-of-its kind" approach to demonstrate additionality and that use versions of baseline and monitoring methodologies, or the "Tool for the demonstration and assessment of additionality" or the "Combined tool to identify the baseline scenario and demonstrate additionality", which allow using the "first-of-its-kind" approach for demonstrating additionality.	Not applicable Project will not be using first of its kind approach for the additionality and therefore this tool is not applicable.

3.3 Project Boundary

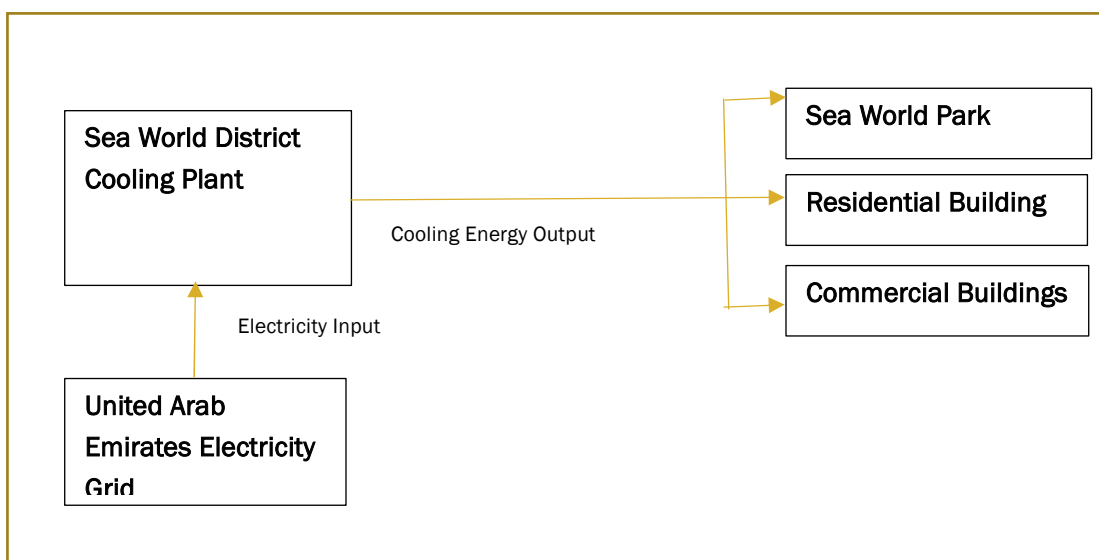
As per the methodology the spatial extent of the project boundary includes:

- (a) Sea World district cooling system, including district cooling plant(s) pipes, sub-stations and buildings, existing and new, that are or will be connected to the district cooling system;
- (b) All the power plant/unit connected physically to the UAE electricity system.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main emission source

Source		Gas	Included?	Justification/Explanation
	Electricity and/or thermal energy consumed by baseline cooling technologies	CH ₄	No	Excluded for simplification, this is conservative
		N ₂ O	No	Excluded for simplification, this is conservative
Project	Electricity and/or thermal energy consumed by the district cooling system	CO ₂	Yes	Main source of emissions
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification

The Project boundary is illustrated in the figure below:



3.4 Baseline Scenario

As per the methodology AM 0117 the baseline scenario is the continuation of the cooling energy production by the baseline cooling technologies. All the conditions are explained in the table below:

Paragraph number/ Condition	Justification
14. The methodology assumes that the baseline scenario is the continuation of the cooling energy production by the baseline cooling technologies.	Applicable In Project the baseline scenario is continuation of the cooling energy production by baseline technologies.
15. District cooling projects implemented in least developed countries (LDCs) are deemed to be automatically additional	Not Applicable UAE is non- LDC and therefore automatic additionality is not applicable.
16. If district cooling projects are implemented in non-LDCs and face the first-if-its-kind barrier, the latest approved version of the “Additionality of first-of-its-kind project activities” shall be followed to demonstrate the additionality of these project activities.	Not Applicable UAE is non – LDC however the project is not first of its kind and therefore this is not applicable.
17. For district cooling projects, which are implemented in non-LDCs and which are not the first-of-its-kind, the procedure described below shall be applied.	Applicable The Project is using this option for additionality and baseline justification.
5.2.1. Step 1: Penetration and performance assessment 18. This step aims to determine whether the proposed project activity is common practice in the city where the proposed CDM project activity is intended to be implemented.	Applicable The Project activity is not a common practice in the UAE which is demonstrated by steps below.
19. The project activity is deemed to be additional if: (a) The seasonal energy efficiency ratio (SEER) of the district cooling plant involved in the project activity (determined as per section 5.2.3.2) is higher than a benchmark SEER (determined as per section 5.2.3.3); and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark	Applicable (a) The SEER (4.84) of Project is Higher than the benchmark SEER (2.26). The detailed description is provided in additionality section below. (b) Share of district cooling technologies in the Abu Dhabi Emirates is 18% which is less than 20% of total cooling output. A third-party survey report is submitted to validator.

boundary in terms of cooling output. The share of technologies can be derived from official country reports, third party surveys and/or credible international sources (e.g. International Energy Agency).	
20. If any of two conditions above is not met, the project proponent should proceed to Step 2.	Not applicable Both the conditions above are meeting.

SEER for Project:

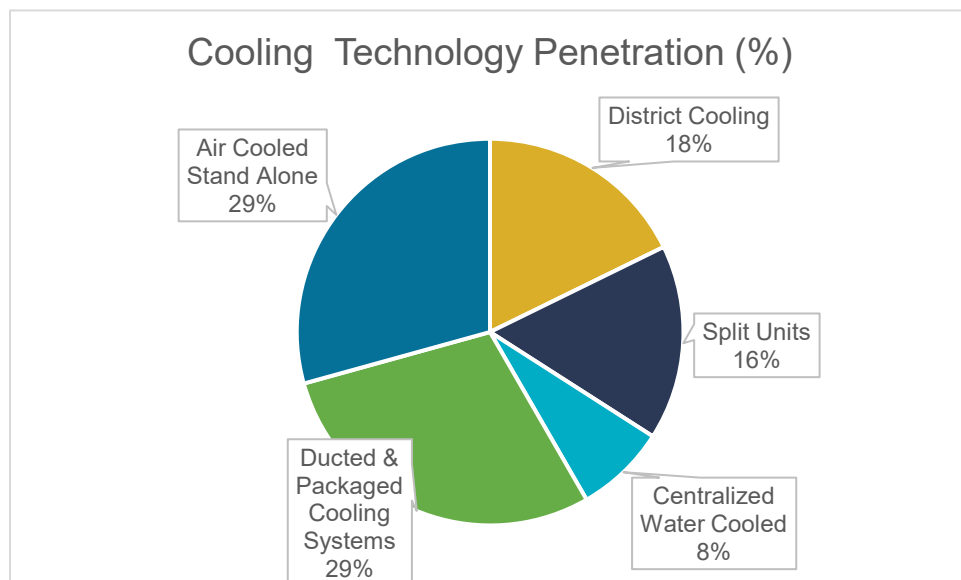
As per the paragraph 19 (a) of the methodology the SEER for the project activity is determined as per section 5.2.3.2.

In accordance with the paragraph 29 of the methodology SEER for the greenfield buildings included in the benchmark boundary shall be determined using only option 2.

Option 2: SEER of the (BAT) in a host country for a building with the same function (e.g. office, apartments) and similar GFA, i.e. in the range from 50 per cent to 150 per cent of the baseline building GFA.

As per the Abu Dhabi Cooling Market report the best available technology in host country is district cooling plant with SEER 4.84 and therefore this is used for project activity.

The share of cooling technologies⁶:



⁶ Abu Dhabi Cooling Market report by GMI (Section 2.6, page 56)

3.5 Additionality

Benchmark determination procedure

The benchmark value serves for two purposes: (a) additionality demonstration and (b) baseline emissions determination.

Benchmark boundary

As per the Methodology the default benchmark boundary is a city where the project activity is located which is Abu Dhabi.

The UAE is divided into seven administrative divisions (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as emirates. Each emirates have its own rules and regulations and clear demarcation of boundary. As no clear administrative demarcation of city is used within the UAE therefore Abu Dhabi Emirate is considered as city for methodological requirement and benchmark boundary. As this is grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates.

The project will not be replacing any existing cooling installations; therefore the benchmark boundary has covered the cooling installations which are providing cooling after installation of district cooling plant. The cooling output of (3,550,000 TR) cooling installation that will continue to provide the cooling is more than the project activity (9000 TR). Therefore, expansion of boundary is not required.

Data vintage and data source

Benchmark SEER shall be determined using SEERs values and cooling output of the installations used within the benchmark boundary i.e. Abu Dhabi Emirate.

The project owner has appointed a third Party⁷ to carry out the Abu Dhabi⁸ cooling sector analysis to provide the benchmark and penetration of colling technologies.

In the benchmark boundary all the operational installations have been considered which can be classified as existing installation. As per the methodology paragraph 26 the new buildings shall also be included in the benchmark boundary which will be included in the future benchmark study report due to unavailability any new buildings in the project boundary.

As per the methodology definitions 'New buildings (sub-area) - buildings within the project boundary that are constructed after the implementation of the project activity and are supplied district cooling by the project activity;'

The project implemented on 25th May 2023 and the benchmark study report has all the data till 31/12/2022 (report received Sep 2023). Therefore, no new building connected after implementation of the project activity.

In the next update of benchmark study report (after 3 years), there will be new building connected with the district cooling and will be included in the benchmark boundary.

⁷ Global Market Insights (GMI) <https://www.gminsights.com/>

⁸ The boundary is ABU Dhabi Emirate

The third party has collected the data from various technology manufacturers, cooling providers, cooling technology users, government entities working in Abu Dhabi and provided independent report. The main data is presented in section below.

The values of SEER for the existing installations included in the benchmark boundary has been determined based on option 1 i.e. Directly from manufacturer of the baseline cooling technology as per the methodology.

The SEER Values of all the installation arrange by type of technology is presented in the table below⁹:

Technology	SEER (BTU/W-Hr)	SEER (MWh/MWh) ¹⁰
District Cooling System	16.5	4.84
Split Units	15.4	4.51
Centralized Water Cooled	14.7	4.31
Others (Ducted & Packaged Cooling Systems)	12	3.52
Air Cooled Stand Alone	7.7	2.26

Benchmark determination

The following steps applied:

- (a) Rank all installations included in the benchmark boundary in order of decreasing SEER;

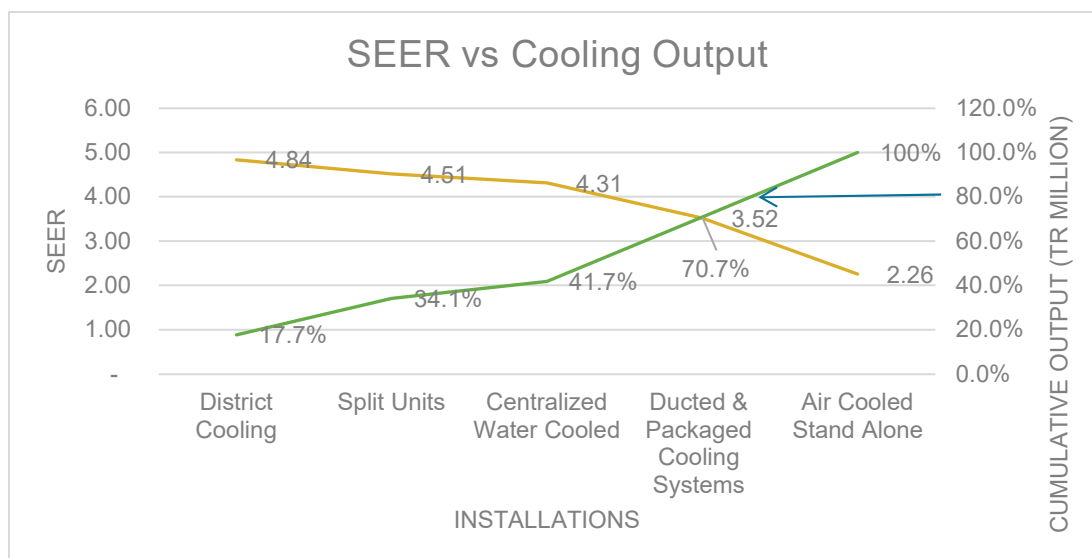
Technology	SEER	Cooling Output in million TR ¹¹	Cumulative Cooling Output (million TR)
District Cooling	4.84	0.63	0.63
Split Units	4.51	0.58	1.21
Centralized Water Cooled	4.31	0.27	1.48
Others (Ducted & Packaged Cooling Systems)	3.52	1.03	2.51
Air Cooled Stand Alone	2.26	1.04	3.55
Total		3.55	3.55

- (b) Plot the SEER values (y axis) as a function of the cumulative cooling output (x-axis);

⁹ Page 56, section 2.6 Abu Dhabi cooling market report by GMI

¹⁰ Conversion is divided BTU/Wh by 3.412 to convert in MWh/MWh

¹¹ Tons of refrigeration



- (c) Identify the benchmark installation that corresponds to at least 80 per cent of the cumulative cooling output;

Benchmark installation at 80% is using Air Cooled Stand alone cooling system.

- (d) Match the value of the benchmark SEER that corresponds to at least 80 per cent of the cumulative cooling output determined in step (c) above.

The SEER value for that installation is 2.26 MWh/MWh

Validity of the benchmark

The first benchmark value shall be identified before the start of the validation process for the purpose of additionality demonstration and ex-ante emission reduction calculations. The benchmark value for baseline emissions estimation is valid for three years.

The project owner will update the benchmark after every 3 years after registration of the project.

The benchmarking report is provided to the validator.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

3.5.2 Additionality Methods

All the steps are explained in the section 3.5 additionality.

3.6 Methodology Deviations

1. Grid emission factor

The methodology refers to CDM Tool 7 for electricity grid emission factor. However, the data is not available in public domain to calculate the grid emission factor and therefore project owner is not able to calculate this factor.

The project owner request to use the publicly available grid emission factor for the project, which will be fix for first crediting period. For future crediting period project owner anticipates that the data will be available to calculate the grid emission factor and then the Tool 7 or any other Verra guidance will be used for the project.

Based on publicly available information, the following data is available for UAE grid emission factor:

Source	Latest year for which emission factor available	Value
Third update for second NDC of the UAE – 2023 page 17 ¹²	2019	0.55 tCO ₂ /MWh
Energy profile, United Arab Emirates published by IRENA – 8 th August 2023 page 3 ¹³	2021	457 tCO ₂ /GWh (0.457 tCO ₂ /MWh)

The Project will consumed less electricity with respect to the baseline and therefore lower grid emission factor will result in lower baseline emissions and lower emission reductions and therefore the lower grid emission factor will be conservative of therefor lower value (IRENA) is selected. As IRENA value is latest and conservative therefore this has been used for the first crediting period.

2. Benchmarking Boundary

¹² https://unfccc.int/sites/default/files/NDC/2023-07/Third%20Update%20of%20Second%20NDC%20for%20the%20UAE_v15.pdf (page 17)

¹³ https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf (page 3)

As per the Methodology the default benchmark boundary is a city where the project activity is located which is Abu Dhabi.

No definition is provided related with the city.

The UAE is divided into seven administrative divisions (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as emirates. Each emirates have its own rules and regulations and clear demarcation of boundary.

- Therefore, Abu Dhabi Emirate is considered as city for methodological requirement and benchmark boundary. As this is grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions are estimated using Approach 1 as the benchmark installation is electricity driven technology.

Total baseline emissions are calculated as follows, using Equation (2) from the methodological tool 05; Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 3.0. When applying the tool, parameter $Q_{B,y}$ shall correspond to parameter $EC_{BL,k,y}$, whereas BE_y shall correspond to $BE_{EC,y}$.

$$BE_y = Q_{B,y} \times EF_{B,y} \times (1 + TDL_{B,y}) \quad \text{Equation (1)}$$

Where

BE_y	=	Baseline emissions in year y (tCO ₂ e/yr)
$Q_{B,y}$	=	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y (MWh/yr)
$TDL_{B,y}$	=	Average technical transmission and distribution losses for providing electricity to the baseline in year y
$EF_{EL,y}$	=	Grid CO ₂ emission factor calculated in accordance with the “Tool to calculate the emission factor for an electricity system” (Version 7.0) (tCO ₂ e/MWh)

BE_y	Baseline emissions in year y	36,393	tCO ₂ e/yr	Calculated
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$Q_{B,y}$	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y	73,614	MWh/yr	Calculated based on Equation 2 below.
$EF_{B,y}$	Grid CO2 emission factor	0.457	tCO ₂ e/MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3
$TDL_{B,y}$	Average technical transmission and distribution losses for providing electricity in baseline in year y	8.18%	%	Department of energy '2021 Annual Technical Report). Page 31 for transmission losses (2.18%) and page 41 for distribution losses (6%)

The estimate cooling output of isolated and less efficiency cooling technologies is determined as follows:

$$Q_{B,y} = \sum_r \frac{OPC_{r,y}}{SEER_{B,i}} \quad \text{Equation (2)}$$

Where:

- $Q_{B,y}$ = Energy consumed in baseline by baseline cooling technologies in year y (MWh)
 $OPC_{r,y}$ = Cooling output of new district cooling plant r in year y (MWh)¹⁴
 $SEER_{B,i}$ = Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)

$Q_{B,y}$	Energy consumed in baseline by baseline cooling technologies in year y (MWh)	73,614	MWh/yr	Calculated
$OPC_{r,y}$	Cooling output of new district cooling plant r in year y (MWh)	166,368	MWh	Estimated based on project capacity (9000 Tons of refrigeration), 8760 hours/year, 60% estimated annual

¹⁴ 1 TR= 3.517 kW = 0.003517 MW

				load factor ¹⁵ , 3.517 is for conversion from TRh to kWh and division by 1000 to convert to MWh
SEER _B	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)	2.26	MWh/MWh	As per the methodology procedure based on data provided by Third party

The grid CO2 emission factor for UAE grid 0.457 tCO2/MWh is used from International Renewable Energy Agency (IRENA) UAE Energy Profile for 2021. This will be fixed for first crediting period.

As per paragraph 39 of the methodology 'The cooling output of the new district cooling plant can be obtained as follows:

Option 1: Direct measurement of the cooling energy (e.g. MWh or GJ), or

Option 2: Calculated based on measurements of temperature differences, flow rate of chilled water and system operating hours per year:

During verification of the project this parameter will be directly measured in Tons hour of refrigeration, which will be converted in MWh. During validation this is calculated based on design data as explained below:

Project cooling capacity = 9000 TR
Annual operating hours = 8760 hr/year
Estimated annual load factor = 60%
Conversion TRh to KWh = 3.517
Cooling output = 9000x8760x0.6x3.517 KWh
= 166368168 kWh
= 166368 MWh

4.2 Project Emissions

Project emissions include emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant.

These emissions are calculated using the "Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" version 3.0.

¹⁵ Based on past operations of other district colling plant, Project is expected to have 60% annual load factor. This is based on project proponents internal assessment.

$$PE_y = PE_{EC,y} + PE_{FC,j,y} \quad \text{Equation (3)}$$

Where:

- PE_y = Project emissions in year y (tCO₂e)
- $PE_{EC,y}$ = Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO₂e)
- $PE_{FC,j,y}$ = CO₂ emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO₂e)

PE_y	=	16994	tCO ₂ /yr	Calculated as per equation 3 above
$PE_{EC,y}$	=	16994	tCO ₂ /yr	Calculated as per equation 4 below
$PE_{FC,j,y}$	=	0	tCO ₂ /yr	This will be zero as project will not use fossil fuel

$$PE_{EC,j,y} = EC_{PJ,j,y} \times EF_{EL,y} (1 + TDL_y) \quad \text{Equation (4)}$$

Where

- $EC_{PJ,j,y}$ = Quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)
- $EF_{EL,y}$ = Grid CO₂ emission factor calculated in accordance with the “Tool to calculate the emission factor for an electricity system” (Version 7.0) (tCO₂e/MWh)
- TDL_y = Average technical transmission and distribution losses for providing electricity in year y

$PE_{EC,y}$	Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y	16,994	tCO ₂ e/yr	Calculated
$EC_{PJ,j,y}$	Quantity of electricity consumed by the new district cooling plant j in year y	34,374	MWh/yr	Calculated based on the cooling load from the project and SEER of district cooling plant (4.84 MWh/MWh). During monitoring actual electricity consumption will be measured.
$EF_{EL,y}$	Grid CO ₂ emission factor	0.457	tCO ₂ /MWh	Energy profile, United Arab Emirates published by IRENA – 8th August 2023 page 3

TDL _y	Average technical transmission and distribution losses in year y	8.18%	%	Department of energy '2021 Annual Technical Report). Page 31 for transmission losses (2.18%) and page 41 for distribution losses (6%)
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$$PE_{FC,j,y} = \sum FC_{i,j,y} \times COEF_{i,y}$$

$PE_{FC,j,y}$	CO ₂ emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO ₂ e)
$FC_{i,j,y}$	Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr) C
$COEF_{i,y}$	Is the CO ₂ emission coefficient of fuel type i in year y (tCO ₂ /mass or volume unit)
i	Are the fuel types combusted in process j during the year y

The project will not be using fossil fuel consumption¹⁶ and will be using only electricity therefore

$$PE_{FC,j,y} = 0$$

4.3 Leakage Emissions

¹⁶ As the project is designed for operation by directly electricity and therefore this parameter will not be monitored during the monitoring period.

Leakage emissions are calculated as follows:

$$LE_y = LE_{Ref,y} + LE_{Ref,sc} + LE_{water} \quad \text{Equation (5)}$$

Where:

LE_y	=	Leakage emissions in year y (tCO _{2e})
$LE_{Ref,y}$	=	Refrigerant leakage emissions from the project in year y (tCO _{2e})
$LE_{Ref,sc}$	=	Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only accounted in the first year of the crediting period) (tCO _{2e})
$LE_{water,y}$	=	Emissions due to the freshwater usage in the project system in year y (tCO _{2e})

LE_y	=	95	tCO ₂ /year	Calculate as per equation 5 above
$LE_{Ref,y}$	=	76	tCO ₂ /year	Calculated as per equation 6 below
$LE_{Ref,sc}$	=	0	tCO ₂ /year	Calculated as per equation 7 below
$LE_{water,y}$	=	19	tCO ₂ /year	Calculated as per equation 8 below

Emissions due to the refrigerants leakage during the project activity is calculated as:

$$LE_{Ref,y} = \sum_k R_{k,y} \times GWP_k \quad \text{Equation (6)}$$

Where:

$LE_{Ref,y}$	=	Refrigerant leakage emissions from the project in year y (tCO ₂)
$R_{k,y}$	=	Quantity of refrigerant k filled in the district cooling system year y (t)
GWP_k	=	Global Warming Potential of the refrigerant k

Type of refrigerant (k)	R 134 A		Actual Data from Chiller Manufacturer's
Global warming potential (GWP _k)	1300	tCO ₂ /ton of R 134 a	AR 5 Report (https://ghgprotocol.org/sites/default/files/ghgp/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf)
Refrigerant charge	25722	Pound (lb)	Actual Data from Chiller Manufacturer's (8574 lb/chiller; total 3 chillers)

	11667.29	kg	Calculated
Estimated annual leakage	0.5 %		Chiller refrigerants: Minimizing carbon footprint, Johnson control (Chiller manufacture) page 5
Quantity of refrigerant k filled in the district cooling system year y (t) $R_{k,y}$	58.34	kg	The project will monitor the actual quantity of refrigerant filled annually. During the validation this is calculated based on annual leakage and total refrigerant use. The leakage quantity will be replenished in the monitoring year
	0.058	tons	Conversion from kg to tons, dividing by 1000.
Emissions ($LE_{ref,y}$)	76	tCO ₂	

Emissions due to the refrigerant leakage from the baseline cooling equipment that is scrapped as a result of the project activity¹⁷ are calculated (only for the first year of the crediting period) as below:

$$LE_{Ref,SC} = \sum_z R_z \times GWP_z \quad \text{Equation (7)}$$

Where:

- $LE_{Ref,SC}$ = Refrigerant leakage emissions from the baseline individual cooling equipment that is scrapped as a result of the project activity (tCO₂e)
- R_z = Quantity of refrigerant z leaked from the baseline individual cooling equipment that is scrapped as a result of the project activity (t)
- GWP_z = Global Warming Potential of the refrigerant z

This is not applicable for project as no baseline equipment are scrapped. Therefore:

$$LE_{Ref,SC} = 0 \text{ tCO}_2/\text{year} \quad \text{Equation (8)}$$

Emissions due to use of freshwater shall be considered in sites where freshwater is produced through the desalination. As emissions associated with the production of make-up water are considered negligible, this source of leakage shall be considered once, i.e. at the first monitoring and calculated as below:

$$LE_{water} = Q_{water} \times EF_{water} \quad \text{Equation (9)}$$

Where:

- LE_{water} = Emissions due to the freshwater usage in the project system (tCO₂e)

¹⁷ Emissions due to the possible refrigerant leakage from the baseline individual cooling equipment that is scrapped as a result of the project activity are accounted taking into consideration the decision taken by the Board at its 34th meeting, Paragraph 17 of the meeting report.

Q_{water} = Maximum designed quantity of freshwater to be used in the project system (t)
 EF_{water} = Emission factor associated with the production of freshwater (tCO₂e/t)

Emissions from fresh water used			Source
Maximum designed quantity of freshwater to be used in the project system (Q_{water})	1620.5	m ³	Design Report page 28 for 9000 TR project activity
Emission Factor (EF_{water})	11.5	kg/m ³	Page 80 Taqa sustainability report 2022
Emissions (LE_{water})	19	tCO ₂	This is applicable for first year only.

The emission factor for water will be provided by water supplier Emirates Water and Electricity Company (EWEC) during the first monitoring. The publicly available value is used for estimation.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (10)}$$

Where:

ER_y = Emissions reductions in year y (t CO₂e)
 BE_y = Baseline emissions in year y (t CO₂e)
 PE_y = Project emissions in year y (t CO₂e)
 LE_y = Leakage emissions in year y (t CO₂e)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
25-May-2023 to 31-December-2023	22,035	10,290	65	11,681	Not applicable	11,681
01-January-2024 to 31-	36,393	16,994	76	19,323		19,323

December-2024						
01-January-2025 to 31-December-2025	36,393	16,994	76	19,323		19,323
01-January-2026 to 31-December-2026	36,393	16,994	76	19,323		19,323
01-January-2027 to 31-December-2027	36,393	16,994	76	19,323		19,323
01-January-2028 to 31-December-2028	36,393	16,994	76	19,323		19,323
01-January-2029 to 31-December-2029	36,393	16,994	76	19,323		19,323
01-January-2030 to 24-May-2030	14,358	6,704	30	7,623		7,623
Total	254,751	118,958	551	135,242		135,242

For projects required to assess permanence risk:

Not applicable

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	Not applicable
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No If no, provide justification.

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO₂e removals and 80,000 tCO₂e reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
DD- MMM- YYYY to 31-Dec- YYYY	Not applicable						
01-Jan- YYYY to 31-Dec- YYYY							
01-Jan- YYYY to DD- MMM- YYYY							
Total							

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	Building Type
Data unit	-
Description	The building types is categorized as new. The project will be providing the cooling to new buildings.
Source of data	<i>District cooling chilled water piping network provided by project owner Tabreed</i>
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	The data is provided by the project owner and checked through the cooling service bills.
Purpose of Data	Calculation of baseline emissions
Comments	-Data will be stored and maintained during verification.

Data / Parameter	Baseline cooling technologies			
Data unit	-			
Description	The data collection has been done by third party based on various data sources, including Abu Dhabi Department of Energy (DoE), Abu Dhabi Industrial Strategy (ADIS), Abu Dhabi Department of Economic Development (ADDED), Abu Dhabi Industrial Development Bureau (IDB) and Global Market Insights (third party consultant). The data from the third party is provided in consolidated form as per the table below due to maintain confidentiality of specific building. The data is segregated based on the baseline cooling technology and type of building. The data set includes from 2019. The independent third party has grouped the baseline cooling technologies and cooling output within Abu Dhabi.			
Source of data	Third Party Survey Report			
Value applied	Technology	SEER	Cooling Output in million TR	Cumulative Cooling Output (million TR)
	District Cooling	4.84	0.63	0.63

	Split Units	4.51	0.58	1.21
	Centralized Water Cooled	4.31	0.27	1.48
	Others (Ducted & Packaged Cooling Systems)	3.52	1.03	2.51
	Air Cooled Stand Alone	2.26	1.04	3.55
	Total		3.55	
Justification of choice of data or description of measurement methods and procedures applied	Data is provided by third party independent company.			
Purpose of Data	Calculation of baseline emissions			
Comments	- Data will be stored and maintained during verification.			

Data / Parameter	EF _{grid,CM,y}
Data unit	t CO ₂ /MWh
Description	Combined margin emission factor for the grid
Source of data	IRENA UAE Energy profile published 8 th August 2023
Value applied	0.457
Justification of choice of data or description of measurement methods and procedures applied	Conservative publicly available information
Purpose of Data	Calculation of baseline and project emissions
Comments	This will be fixed for first crediting period.

Data / Parameter	Q _{water}
Data unit	Ton

Description	Maximum designed quantity of freshwater to be used in the project system
Source of data	Concept Design Report Page 28 for 9000 TR
Value applied	1620.52
Justification of choice of data or description of measurement methods and procedures applied	Design data
Purpose of Data	Calculation of leakage
Comments	This data will be used for first year only as per the methodology.

5.2 Data and Parameters Monitored

Data / Parameter	OPC,r,y
Data unit	TRh
Description	Cooling output of new district cooling plant r in year y
Source of data	Data is based on <i>cooling output</i> meters recorded in AD026-Daily Profile sheet.
Description of measurement methods and procedures to be applied	<i>This will be monitored and calculated through online monitoring system.</i>
Frequency of monitoring/recording	Continuous
Value applied	47304000 (TRh) 166368 (MWh)
Monitoring equipment	<i>Cooling Output Meters</i>
QA/QC procedures to be applied	Meter Serial number: T4045F043BC and T4045F043BC Calibration frequency: Annual Cross check procedure: The cooling output meters are installed at Energy Transfer Stations. The monthly data will be consolidated from all the energy transfer stations (ETS)

	connected with the district cooling plant. The lower value between consolidated ETS data and District cooling plant will be considered (conservative). This will be crosschecked with invoice as well which will be issued to the user of cooling. Most conservative value will be used for emission reduction.
Purpose of data	Calculation of baseline emissions
Calculation method	As per equation in Baseline Emission section above. The TRh value will be converted to MWh by multiplication of conversion factor 0.003517
Comments	- For estimation 60% operating capacity included. Project capacity 9000 TR, 8760 hrs/year, 60% operating capacity =47304000 TRh

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh
Description	Quantity of electricity consumed by the new district cooling plant j in year y
Source of data	Monthly ADDC Invoice
Description of measurement methods and procedures to be applied	This will be monitored and calculated (addition of the electricity data as two meters are available) through online monitoring system.
Frequency of monitoring/recording	Continuous
Value applied	34,374
Monitoring equipment	Electricity Meters
QA/QC procedures to be applied	Meter number: E2T12D000237 and E2T12D000237 Calibration: Annual Cross check procedure: The power monitoring units are installed parallel to ADDC meters at power incomers. The values are recorded daily at the site office. This monthly value will be crosschecked with ADDC invoice reading. In case of any

	differences higher value (conservative) will be considered in emission reduction calculation.
Purpose of data	Calculation of project emissions
Calculation method	<i>KWh value will be converted in MWh by dividing by 1000.</i>
Comments	-The value is calculated based on the district cooling SEER by third party (4.84 MWh/MWh). During monitoring actual value will be monitored.

Data / Parameter	TDL _{B,y} and TDL _y
Data unit	%
Description	Average technical transmission and distribution losses for providing electricity in year y
Source of data	Annual average value based on the most recent data available within the host country;
Description of measurement methods and procedures to be applied	This will be based on published reports.
Frequency of monitoring/recording	Annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years
Value applied	Transmission loss: 2.18% Distribution loss: 6%
Monitoring equipment	- Not applicable
QA/QC procedures to be applied	Using credible data for electricity system (default value)
Purpose of data	Calculation of baseline and project emissions
Calculation method	.Transmission loss and distribution loss values will be added and used in the equation.

Comments	-Data from Department of Energy's technical report 2021 is used during validation.
Data / Parameter	SEER _{B,i}
Data unit	MWh/MWh
Description	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i
Source of data	From manufacturers of the baseline cooling technology
Description of measurement methods and procedures to be applied	The value will be provided by independent entity.
Frequency of monitoring/recording	Every 3 years
Value applied	2.26
Monitoring equipment	Not applicable. Based on third party report.
QA/QC procedures to be applied	- Based on third party report
Purpose of data	Calculation of baseline emissions
Calculation method	<i>Units will be converted as per the report provided. Based on the present report BTU/W.h is converted to MWh/MWh by dividing it to 3.412.</i>
Comments	- this will be valid for 3 years baseline emission calculations.

Data / Parameter	R _{k,y}
Data unit	T
Description	Quantity of refrigerant k used in the project in year y
Source of data	Records from the plant operator
Description of measurement methods	Metering will rely on the simple counting of cylinders

and procedures to be applied	
Frequency of monitoring/recording	Continuous, As and when cylinder will be used
Value applied	0.058
Monitoring equipment	Supplier
QA/QC procedures to be applied	Crosschecked with purchase records. All meters and scales will be calibrated as per manufacturers' recommendations
Purpose of data	Calculation of leakage emissions
Calculation method	- The total quantity of refrigerant used to fill the chillers in the project activity will be added.
Comments	- DOE will validate this data.

Data / Parameter	GWPk,
Data unit	- unitless (its ratio)
Description	Global Warming Potential of the refrigerant k
Source of data	Records from the plant operator
Description of measurement methods and procedures to be applied	Intergovernmental Panel of Climate Change (IPCC)'s latest reports or any other relevant scientific body's assessment
Frequency of monitoring/recording	-As and when new IPCC report published with updated global warming potential values.
Value applied	1300
Monitoring equipment	- Not applicable
QA/QC procedures to be applied	- This is from authentic sources/default value

Purpose of data	Calculation of leakage emissions
Calculation method	- Not applicable
Comments	- Shall be updated according to any future COP/MOP decisions

Data / Parameter	EF _{water}
Data unit	tCO _{2e} /t
Description	Emission factor associated with the production of fresh water
Source of data	Emirates water and electricity company
Description of measurement methods and procedures to be applied	-this data is from official source.
Frequency of monitoring/recording	-First monitoring period
Value applied	0.0115
Monitoring equipment	- Not applicable. Data is provided by third party.
QA/QC procedures to be applied	- This is data from the water producer which is supplier to full Abu Dhabi and will follow its internal QA/QC process
Purpose of data	Calculation of leakage emissions
Calculation method	- Not Applicable
Comments	-The publicly available value is used during validation.

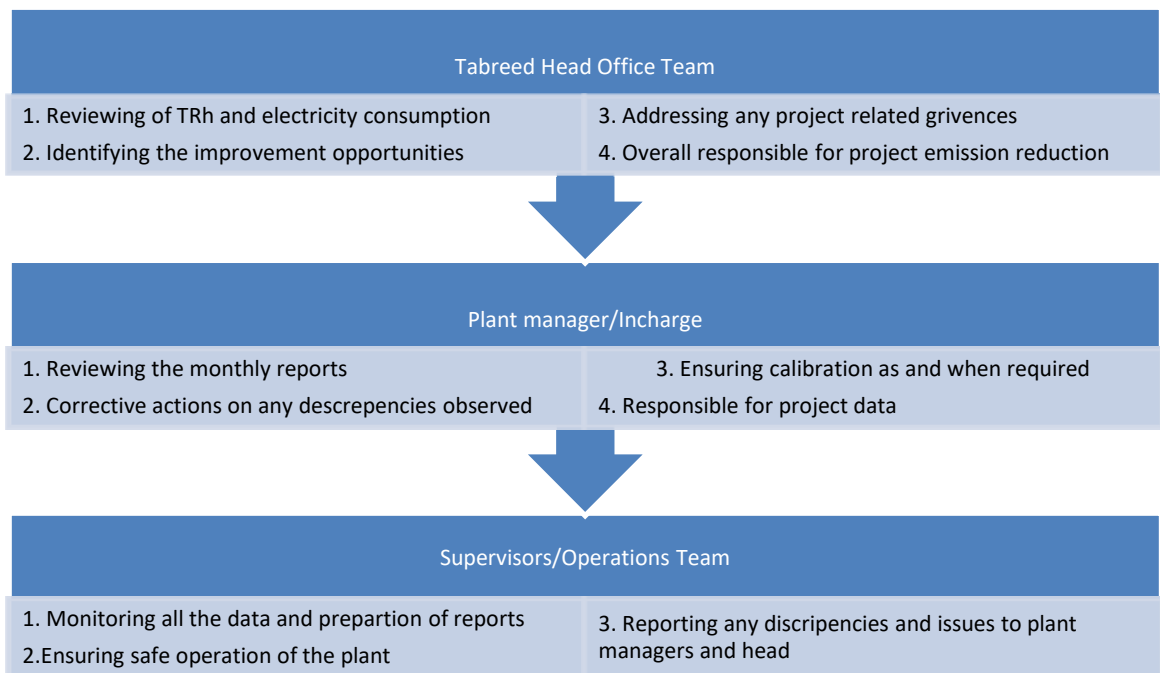
5.3 Monitoring Plan

The project is operated by Tabreed which ensures the overall site management in accordance with the UAE Laws and technology providers' guidelines and will perform the monitoring in house.

The project monitoring will comply with the monitoring methodology AMO117 Version 2.0. Therefore, all relevant parameters stated in Section 5.1. above will be reliably monitored and cross checked through calibrated measurement equipment.

Additionally, all workers will receive the appropriate training for the operation and maintenance of all equipment based on the monitoring equipment maintenance/calibration procedures, the emergency strategy and sampling protocols for waste composition analysis.

All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the last crediting period.



APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
	Not applicable	